

Isaac Au

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EDUCATION

Carnegie Mellon University

B.S. in Computer Science, **GPA: 3.9**, Dean's List High Honors (2023 - 2026)

Pittsburgh, PA

Aug 2023 – May 2027

Relevant Coursework: ML for Large Datasets, Deep Learning, Computer Vision (Grad), Advanced NLP (Grad)

WORK & RESEARCH EXPERIENCE

Google | *Software Engineer (incoming)*

Sunnyvale, CA | Starting Jun 2027

- AI Catalyst Program in AI and Infrastructure (AI2) Organization.

TDK | *Machine Learning Intern*

Pittsburgh, PA | May 2026 – Aug 2026

- Built an LLM-guided evolutionary search agent (modeled on DeepMind's AlphaEvolve) that Pareto-optimizes tinyML models across competing metrics like accuracy vs. latency; adopted by other engineers via CLI configs, experiment tracking, and parallel serverless evaluation (Bedrock, Lambda/ECR/S3).
- Cut LLM cost ~30% and generation time ~20% with no failure-rate increase by replacing a ~1000-line in-context evaluator with an LLM-summarized contract, validated via 3-arm ablation.
- Reduced a production signal reconstruction algorithm's latency ~98%, and surfaced a wearable crash-detector variant at ~50% ROM within 0.8 pt of baseline accuracy (~\$20, ~1 hr per run).

Ovis Medical Solutions | *Co-founder, Technical Lead*

Hong Kong | Feb 2024 – Feb 2026

- Shipped an agentic oncology clinical assistant with function calling, privacy-screened long-term memory across check-ins, and guardrailed tool execution, cutting patient wait from ~30 s to under 1 s via async triage.
- Built policy-as-code LLM routing keeping patient chats self-hosted and sending de-identified transcripts to Azure OpenAI, behind a PHI guardrail ~12× faster and more precise than default Presidio on Cantonese text.
- Secured \$100,000 in SAFE funding, led a 5-person team and collaborated with University of Hong Kong oncologists through the full-stack MVP lifecycle and IRB submission for a clinical pilot.

CMU 15-440 Distributed Systems | *Teaching Assistant*

Pittsburgh, PA | Aug 2026 – Present

- Lead weekly recitation for 40 students and hold office hours across the course's systems projects: a key-value server, a replicated distributed store, and a distributed worker pool; author exam questions.
- Building experimental AI course tooling: a harness that generates personalized quiz questions from project submissions, and opt-in instrumentation to study student AI-agent usage patterns.

Research in Medical LLM Data and Evaluation | *with Prof. Bryan Wilder*

Pittsburgh, PA | May 2025 – Sept 2025

- Built an LLM-as-a-judge pipeline filtering 2M+ clinical entries into a 50k-row instruction and thinking dataset.
- Ran LoRA-based supervised fine-tuning via LlamaFactory across Qwen foundation models, improving medical-reasoning accuracy ~15% on a curated multiple-choice benchmark of complex cases.

Rice Robotics | *Software Development Intern*

Hong Kong | May 2024 – Jul 2024

- Built an Azure OpenAI autonomy system for an embodied AI robot, cutting round-trip latency 3.5× (~10s → ~3s) by multithreading the multimodal pipeline and parallelizing sentiment analysis, TTS, and motor-command generation against streamed output tokens.

PROJECTS

DropIn — Real-Time IMU Pose Estimation on Consumer Devices

Feb 2026 – Jul 2026

- Engineered a real-time markerless pose estimation system reconstructing full-body kinematics from iOS IMU streams, using quaternion-based inverse kinematics as a low-cost alternative to \$50k+ optical mocap setups.
- Decoupled 100Hz dual-IMU sensors from a fixed-rate renderer with a conflating latest-value buffer, and guarded reconnect and client-state resync, keeping render latency bounded under unreliable WiFi network.

GUI-Grounded Generation

Mar 2026 – May 2026

- Built a zero-shot inference pipeline injecting frozen GUI-grounding models for computer-use agents (OmniParser v2, JEDI) as structured prompt scaffolding into a VLM code-repair model without fine-tuning.
- Drove a +29% gain in visual fidelity (CLIP, $p < 0.01$) on Angular tasks for DesignBench's 111-sample repair task.

SKILLS

Programming Languages: Python, C, C++, Go (Golang), SQL, Bash/Shell, Swift

Infrastructure: AWS (Bedrock, Lambda, ECR, S3), Docker, PostgreSQL, MongoDB, FastAPI, Vue, Git, Linux

ML/AI: PyTorch, Hugging Face, Polars, Weights & Biases, Ollama, FAISS (RAG), LlamaFactory, LLM-as-a-judge

Languages: Fluent in English, Cantonese, Mandarin Chinese